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Real-time Mitigation of LiDAR Mutual Interference

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***Abstract*— Future automated vehicles are likely to rely on LiDAR sensors to perform automated functions. These LiDAR sensors have the potential to interfere with each other, with this mutual interference being represented by unwanted points in the point cloud. These points could affect the quality of perception of objects, such as potentially creating false objects or obscuring real objects. Hence, it is necessary to identify interference points and remove them from the point cloud, without removing points that belong to valid objects. This paper presents a novel LiDAR mutual interference mitigation algorithm that successfully mitigates interference and potentially can be performed in real time. The number of interference points when the victim and offending LiDARs were separated by 1m was reduced from over 1000 points to under 40 points. The proposed method performed better than the current state-of-the-art techniques on most metrics and was designed to ensure that consecutive points at the same range, such as those belonging to small objects at far distances, were not permanently removed from consecutive point clouds. These results ensure a removal of interference point without affecting the accuracy of perception tasks based on LiDAR data.**

I. Introduction

Automated vehicles rely on sensors to perceive their surrounding environment, such as camera, RADAR, and LiDAR sensors [1]. As the prevalence of automated technology increases, there is the potential that a greater number of vehicles will be fitted with LiDAR sensors [2]. Hence, there will likely be more LiDAR sensors interacting in the same environments in close proximity. Since LiDAR sensors emit infrared beams into the environment, they could be received by LiDAR sensors equipped to other vehicles [3]. This could cause erroneous detections, which have the potential to affect the vehicle's ability to perceive the environment [3].

Other automotive perception sensors are also known to have weaknesses. For example, camera sensors are negatively affected by low-light conditions, and RADAR sensors have poor resolution [4], [5]. Hence, LiDAR sensors can complement camera and RADAR by providing high spatial resolution regardless of light conditions [6]. High resolution is particularly important for dense urban environments where detecting Vulnerable Road Users (VRUs), such as pedestrians and cyclists, is of great importance. However, if the LiDAR sensor is impacted by mutual interference, its performance at detecting VRUs will likely be reduced. Therefore, LiDAR mutual interference mitigation is essential.

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Some of the current proposed mitigation techniques either work on a single point cloud, or have an inherent delay [3], [7]. However, not using historical point cloud data risks dismissing small or hard to detect objects as interference. Furthermore, an inherent delay means the mitigation can never be truly real-time.

The proposed LiDAR mutual interference method considers point clouds from the two previous point clouds and does not have an inherent delay. It is shown to be more effective at mitigating interference than established methods, and can be applied to point clouds in real-time. The proposed method mitigates the strongest interference within just a few point clouds and maintains strong accuracy against baseline point clouds over time.

II. Background

A. Measured Mutual Interference in LiDARs

There has been some previous work completed measuring and reporting on LiDAR mutual interference. Diehm et al. observed over 1000 (1.5%) interference points between two Velodyne HDL-64E LiDAR sensors [3]. Eom et al. found mutual interference present in SICK LMS-511 LiDAR sensors, with thousands occurring in consecutive scan lines [8]. Popko et al. found 1.04% of points in the worst case were interference points using SICK LMS-531 LiDAR sensors [9]. Nunes-Pereira et al. found that the Garmin Lite v3 interfered with a Quanergy M8 but not vice versa, and that the interference points were random in range [10]. Briñón-Arranz et al. experimented with five anonymised LiDAR sensors, discovering that in 6 out of 8 cases, the victim LiDAR experienced a reduction in returns - and a large variation in range returns in some victim LiDARs - from interference mutual interference [11]. Robinson et al., on the other hand, found little evidence of mutual interference when using Ouster OS1-64 as the victim LiDAR [12]. Alam and Rajalakshmi found no evidence of interference in three Velodyne LiDARs, leading them to suggest that the manufacturer had mitigated interference internally [13].

Hence, LiDARs have been shown to be susceptible to interference from other LiDARs, but part of this work has been conducted using LiDARs not representative of the current state-of-the-art. More recent work has shown that some LiDARs appear to be resistant to interference from other LiDARs. To the Author's knowledge there has not been any work measuring interference between two non-scanning Micro-Electro-Mechanical Scanning (MEMS) LiDARs, a technology prevalent on the market in 2024. Hence, this work collected data using this type of LiDAR.

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B. Mutual Interference Mitigation Methods

This section focuses on mitigation methods that aim to remove interference from the LiDAR point cloud. There are also works looking at integrating interference resistance into the LiDAR such that it does not suffer from interference [14], [15]. However, as shown in Section II-A, LiDAR mutual interference is still present in the latest LiDAR sensors and hence work to mitigate interference in the point cloud is still needed.

Han et al. presented a comprehensive review of point cloud filtering methods, categorised into different types of filters [16]. These categories include: Statistical-based filtering techniques, Neighbourhood-based filtering techniques, Projection-based filtering approaches, Signal Processing-based methods, PDE-based filtering techniques, Hybrid filtering techniques and Other methods. More point cloud filtering methods can be found in the Point Cloud Library (PCL), including the C++ code for some of the example filtering techniques presented in Han et al. [16], [17].

Most of the techniques presented in Han et al. are applied on a single point cloud to remove noise, often in the context of point clouds representing an object rather than a scene [16]. This means they are not directly transferable to the case of a LiDAR scan of an environmental scene with mutual interference. Interference is often witnessed in varying positions and can be difficult to distinguish from small objects at long distances, which can be represented by few sparse points in a point cloud. As a result, while operating on a single point cloud may be successful at removing interference, it could also remove points belonging to small objects at longer distances. Hence, temporal information must also be used.

The Statistical outlier removal method, the Spatio-temporal method from Diehm et al. and the Kalman filter were chosen for further exploration in this paper [3], [7], [18]. The Statistical outlier removal filtering first processes the point cloud to find average and standard deviation of k nearest neighbours to each point and uses these with a multiplier to define the threshold. The point cloud is then reprocessed to remove points whose average distance to its k-nearest-neighbours is above or below this threshold. This method was chosen for its combination of statistical and neighbourhood-based techniques, which effectively isolate outliers such as those caused by interference in point clouds generated by LiDAR sensors.

The Spatio-temporal method proposed by Diehm et al. mitigates interference by comparing points in each time step with the previous and next time step [3]. Their method uses a nearest-neighbour algorithm to find the nearest point in the previous and next point cloud. If this point is beyond a threshold distance from its nearest-neighbours it is removed. This method considers more than one point cloud but an inherent delay of a single time step is introduced since a future point cloud cannot be used.

The Kalman filter is a well-known filtering algorithm for removing noise and converging to a true state [18]. Whilst this is usually used to smooth noise and provide accurate system states even in the absence of measurements over time, there is a potential application to temporally aligned point clouds. The benefit of the Kalman filter - if it is shown to be accurate - is that it is easily extendable to dynamic scenarios.

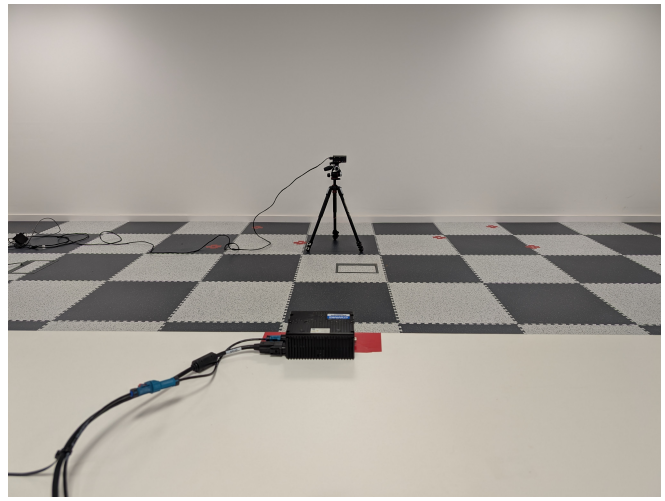

Fig. 1. Test setup used with the offending LiDAR positioned 3m from the victim LiDAR.

This work presents a method that uses temporally historic point clouds without introducing an inherent delay. The proposed method is compared against the Statistical outlier removal method, the Spatio-temporal method from Diehm et al. and the Kalman filter to assess its performance.

III. Methodology

A. Data Collection

Two MEMS LiDAR sensors from different manufacturers were used for collecting data to test for mutual interference. One LiDAR was used as the victim, the other was used as the offender. The victim LiDAR was positioned facing a clean wall at a distance of approximately 4m. The offending LiDAR was placed in front and facing the victim LiDAR, at distances of 1m, 2m, and 3m , using a laser pointer to ensure distances were accurate to within 5mm. The offender was at approximately the same height as the victim. An image of the experimental setup is shown in Fig. 1.

Approximately 20s of data was collected with the offending LiDAR turned off to provide a baseline measurement and then with the offending LiDAR turned on to test for interference. This was repeated twice to give 3 sets of 20s of data for each test. The results given in Table I have been averaged across these 3 tests.

B. Proposed Mitigation Method

The newly developed LiDAR mutual interference mitigation method aims to remove interference points whilst keeping any valid points that may belong to smaller objects at longer distances. To accomplish this, the method should consider historical data, including previously removed points. This is to ensure that if a point appears in the same angle at a very similar range in consecutive point clouds, it is not removed because it could be a valid object point. The methods

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should also not have any inherent delay and should consider more than a single point cloud. Our method balances the need to remove interference with the desire to avoid removing points that potentially belong to real detections.

The proposed LiDAR mutual interference method is a neighbourhood-based method and uses the nearest-neighbour search to find the closest point in historical point cloud data to any given point in the current point cloud, PC_t. For each point in the current point cloud, the nearest point in azimuth and elevation is found in PC_{t-1} and PC_{t-2}. Range is not used, to try and avoid situations where a cluster of interference points is not removed due to their proximity to each other. The points are then compared in range. If the difference in the range of the current point and nearest points in PC_{t-1} and PC_{t-2} is greater than a specific range threshold, then the point is removed. If one or more of the nearest historical points are within the range threshold of the current point, it remains. This range threshold has been varied for the analysis with results presented using range thresholds of 0.5m, 0.75m and 1.0m.

However, a memory of the removed points is kept for use in ensuring that points that belong to real detections are not removed. For each point in PC_t, the nearest point in the set of removed points from $t-1$ and $t-2$ is found and the difference in range compared to the range threshold. If a candidate for removal has range within the range threshold of both of the nearest points in the set of removed points in $t-1$ and $t-2$, then it is kept. This ensures that if a new object is detected by the LiDAR sensor and points appear in the point cloud representing that object, it is not removed from consecutive point clouds. It would instead be re-added after a maximum of two time steps.

C. Methods for benchmarking

This proposed method has been compared against the methods described in Section II-B to test its performance, namely the Statistical outlier removal method (implemented in MATLAB using pcdenoise function), the Kalman filter, and the Spatio-temporal method from Diehm et al. [3]. Statistical outlier removal method has been chosen due to its relative simplicity of computation as it works on each point cloud independently. Kalman filter has been chosen because of its prominence in filtering out noise, examining its suitability for removing LiDAR mutual interference. The method from Diehm et al. allows comparison against a mitigation method that also uses historical temporal data to remove interference [3]. Mitigation was conducted from the third point cloud in the dataset due to the need for the proposed method to use two historical point clouds. This delay will only be present on startup of the sensor. Due to the need for a $t+1$ point cloud, the Diehm method was applied to one less point cloud.

The pcdenoise function and Diehm method were applied to each point cloud consecutively and required no adjustment. The Kalman filter needed adjustments to the input data to be used for mitigation. Since the data only contained values for points returned, the size of each point cloud varied frame by frame and was not consistent. Therefore, point clouds were reconfigured into a matrix with azimuth in one dimension, elevation in the other, and the range stored as the value. Precision for azimuth and elevation bins was chosen to be $0.1°$ per bin as per the LiDARs resolution. Storing by angle resulted in some angle bins regularly containing a zero value for range, causing both estimate error and measurement error to be zero. Kalman gain was forcibly set to 1 in these instances, to avoid it being undefined. The Kalman Gain was initialised at 0.5. The Kalman filter could then be applied on these matrices to mitigate interference.

D. Metrics

The selected metrics for assessing success of the point cloud mitigation algorithm are the number of remaining interference points, the Hausdorff distance and the computation time. The number of remaining interference points gives a direct understanding of the success of removing points. The Hausdorff distance shows how close the mitigated point cloud is to the one collected without interference, showing where mitigation creates distortion of valid points. The computation time is important to assess as these methods will have to run in real time in practice. In this work we have not optimised the implementation, instead we have used MATLAB for all and included the computation time for benchmarking purposes.

1) Number of Interference Points: The main metric used for assessing whether a LiDAR has been affected by mutual interference is the number of remaining interference points. To assess whether a point is an interference point, the baseline point cloud (with offender turned off) was first filtered to remove noise by taking only points with range less than 10m. This is possible in this static scenario we know that the real distances are less than 10m. Then the average and standard deviation of range in each angle was calculated. Angle bins of $0.1°$ are used to group points that belong to the same LiDAR beam. For the test case with offending LiDAR turned on, the associated angle bin is determined and the range compared to the average found in the baseline point cloud. If the difference is greater than the specific range threshold, and 3 times the standard deviation for that azimuth and elevation, the point is classed as an interference point. The number of interference points was calculated for each mitigation method and each separation distance, with range thresholds of 0.5m, 0.75m and 1.0m.

2) Hausdorff Distance (HD): The bilateral Hausdorff distance has been used to determine how close the mitigated point cloud is to the baseline point cloud. For each point in the reference point cloud, the distance to the closest point in the target point cloud is found and the maximum of these distances is taken. For the bilateral Hausdorff distance, the process is repeated with the point clouds swapped.

$$HD = \max \begin{pmatrix} \max_{p_r \in PC_{ref}}(\min_{p_t \in PC_{tar}}(d(p_r, p_t))), \\ \max_{p_t \in PC_{tar}}(\min_{p_r \in PC_{ref}}(d(p_r, p_t))) \end{pmatrix} \quad (1)$$

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3) Hausdorff 95% *Distance:* Hausdorff 95% Distance was considered but omitted from the final results. Since there are less than 5% interference points in the data, by using Hausdorff 95% distance, only the expected non-interference points are considered.

4) Computation Time: Total computation time has been measured and divided by the number of point clouds to give the computation time per frame. Any manipulation or formatting of the data has been ignored and only the implementation of the mitigation has been timed. All the methods have been implemented in MATLAB and tested using a laptop with a 1.8GHz processor and 16GB RAM. The times are presented for comparison with each other rather than real-time, as a different implementation or compute would likely see improvements to all the presented methods.

IV. Results

A. Measured Interference

For the original data without mitigation, the number of interference points averaged across the 3 data collections is shown in Table I. The minimum, maximum, and mean number of interference points across the point clouds is given for each test setup. The results are shown for three range thresholds: 0.5m, 0.75m and 1.0m. There was a mean of around 650 interference points in each point cloud with 1m separation, around 170 interference points with 2m separation and 95-100 interference points with 3m separation for all three range thresholds. Applying this metric to the data collected with the offender turned off, the maximum number of "interference" points in each point cloud averaged across the 3 data collections was 15. This represents the noise points that our metric classifies as interference. Hence, it is clear that our data contains mutual interference points as is emphasised by the point clouds shown in Fig. 2.

B. Mitigation

The results for the minimum, maximum and mean number of interference points for each mitigation method are also shown in Table I, along with the minimum, maximum and mean Hausdorff distances and computation time. The best performing method for each separation distance, range threshold and metric combination is shown by blue shading of the cell.

For the proposed method, the choice of range threshold had limited effect on the number of interference points, Hausdorff distance and the computation time. For Diehm, there was limited effect except for the number of interference points, which performed better with a range threshold of 0.5m for 1m separation. Kalman filter and and pcdenoise do not use the range threshold and hence were unaffected. In 7 out of 9 conditions the proposed method performed best of the compared mitigation methods for maximum and mean number of interference points and minimum and maximum Hausdorff distance. The proposed method performed best on these four metrics for all the cases with 1m separation where interference was strongest.

The Kalman filter performed best for minimum number of interference points and worst for maximum number of interference points for all conditions. The Kalman filter also performed best for mean Hausdorff distance at 1m and 3m separation for all conditions. Table II shows the results for frames 3 to 10. Frame 3 was the first point cloud that was mitigated as described in Section III-C. Over these frames, the proposed method performed best for all metrics. Kalman filter introduced noise over the first few mitigated frames, showing that whilst over a long period of time the Kalman filter was very successful at removing interference, it struggled over a shorter period of time. It even created more interference points initially before successfully mitigating over time. The proposed method had a maximum of 32 interference points over frames 3 to 10, indicating successful mitigation from the first point cloud.

The Kalman filter was by far the most computationally efficient method. The proposed mitigation method and the pcdenoise (Statistical outlier removal) method were somewhat comparable in computation time with the pcdenoise slightly faster. Diehm method took the longest time to mitigate each point cloud.

V. Discussion

The results presented above show that interference was observed in the victim LiDAR with more than 1000 interference points in some point clouds when the LiDARs were separated by 1m. Even with 3m separation, over 200 points in some cases were interference points. This is a substantial number of points that could potentially cause problems for perception tasks.

The mitigation method proposed in this paper performed the best at removing interference points from the point cloud, achieving the lowest maximum and mean number of interference points for all range thresholds and separation distances, with only the Kalman filter performing better for minimum number of interference points. For the Hausdorff distance, the results were more mixed. The proposed method performed best for maximum Hausdorff distance in all conditions. However, the Kalman filter had the best mean Hausdorff distance at 1m and 3m separation, and the pcdenoise performed the best for minimum Hausdorff distance in some cases, albeit only just.

Combining these metrics builds a better picture, as the Kalman filter had thousands of interference points in some point clouds, and the pcdenoise had hundreds. In some cases, the Kalman filter had more "interference points" than the original non-mitigated point cloud. This is because the "interference points" are not removed but moved to be closer to the valid range. Initially this adds noise to the point cloud, before the filter returns the point to the expected range over time. This effect can be seen visually for frame 10 in Fig. 2. Since the offending LiDARs are to be fitted to moving vehicles, the interference points are going to be at their strongest for a short period of time and their position in the victim point cloud will change over a longer period

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TABLE I

PERFORMANCE OF EACH LIDAR INTERFERENCE MITIGATION METHOD FOR DIFFERENT SEPARATION DISTANCES WITH DIFFERENT RANGE THRESHOLDS. BLUE CELLS REPRESENT THE BEST PERFORMANCE FOR EACH METRIC IN EACH EXPERIMENT.

Range Threshold	Distance	Mitigation Method	Interference Points			Hausdorff Distance (m)			Computation Time
			Min	Max	Mean	Min	Max	Mean	
0.5m	1m	No Mitigation	70	1099	652	165.96	238.32	221.40	N/A
		Proposed	7	34	19	0.13	176.07	38.89	0.786
		Diehm	33	247	118	92.09	227.77	197.79	1.040
		Kalman	0	20303	583	0.16	220.90	25.73	0.235
		pcdenoise	26	964	545	28.5	233.18	188.40	0.547
	2m	No Mitigation	33	341	172	117.28	229.91	203.23	N/A
		Proposed	10	48	24	0.14	177.24	43.57	0.778
		Diehm	30	155	54	113.04	214.85	170.95	1.115
		Kalman	0	20385	599	0.16	214.86	50.91	0.253
		pcdenoise	20	304	144	0.14	224.54	153.12	0.604
	3m	No Mitigation	26	204	98	113.55	224.08	197.47	N/A
		Proposed	7	42	21	0.10	177.24	40.43	0.784
		Diehm	21	70	40	93.01	217.03	170.40	1.088
		Kalman	0	20254	580	0.14	197.72	24.43	0.237
		pcdenoise	13	173	78	0.11	223.07	145.08	0.575
0.75m	1m	No Mitigation	70	1098	651	165.96	238.32	221.40	N/A
		Proposed	6	35	18	0.13	184.54	46.91	0.770
		Diehm	38	325	151	112.03	230.40	203.29	1.003
		Kalman	0	20303	570	0.17	220.91	26.58	0.219
		pcdenoise	25	962	543	28.5	233.18	188.40	0.534
	2m	No Mitigation	31	333	168	117.28	229.91	203.23	N/A
		Proposed	8	45	23	0.13	177.39	48.15	0.823
		Diehm	28	153	53	113.04	214.85	173.06	1.078
		Kalman	0	20382	559	0.16	214.86	49.52	0.241
		pcdenoise	17	294	139	0.14	224.54	153.12	0.559
	3m	No Mitigation	24	202	96	113.55	224.08	197.47	N/A
		Proposed	8	43	20	0.12	177.55	55.12	0.773
		Diehm	21	67	39	93.66	218.18	171.84	1.004
		Kalman	0	20253	546	0.14	197.70	24.66	0.219
		pcdenoise	10	170	76	0.11	223.07	145.08	0.537
1.0m	1m	No Mitigation	70	1098	650	165.96	238.32	221.40	N/A
		Proposed	6	36	18	0.14	184.54	47.43	0.756
		Diehm	43	393	183	112.14	230.40	206.81	1.006
		Kalman	0	20303	546	0.17	220.91	29.38	0.227
		pcdenoise	25	962	542	28.5	233.18	188.40	0.560
	2m	No Mitigation	31	333	167	117.28	229.91	203.23	N/A
		Proposed	8	45	22	0.12	186.63	50.25	0.790
		Diehm	29	160	55	113.04	214.85	174.03	1.065
		Kalman	0	20382	551	0.16	214.86	53.48	0.236
		pcdenoise	16	293	138	0.14	224.54	153.12	0.575
	3m	No Mitigation	23	201	96	113.55	224.08	197.47	N/A
		Proposed	8	45	20	0.12	177.55	60.90	0.737
		Diehm	22	70	40	93.73	218.18	172.07	1.013
		Kalman	0	20253	561	0.14	197.71	29.02	0.220
		pcdenoise	9	170	76	0.11	223.07	145.08	0.528

TABLE II

PERFORMANCE OF EACH LIDAR INTERFERENCE MITIGATION METHOD FOR 1M SEPARATION DISTANCES WITH RANGE THRESHOLD OF 0.5M OVER FRAMES 3 TO 10. BLUE CELLS REPRESENT THE BEST PERFORMANCE FOR EACH METRIC IN EACH EXPERIMENT.

Mitigation Method	Interference Points			Hausdorff Distance (m)		
	Min	Max	Mean	Min	Max	Mean
Proposed	11	28	19	0.17	175.94	66.19
Diehm	43	185	108	171.58	222.20	202.57
Kalman	1187	20303	5995	92.54	220.90	180.29
pcdenoise	48	892	505	104.53	227.09	192.46

of time. Therefore, the proposed method is more suitable for mitigating interference than the Kalman filter.

The proposed method's performance was at its strongest in comparison to the other mitigation methods at 1m, indicating that it is better at dealing with large amounts of interference. This suggests a strong performance potential when there are multiple offending LiDARs operating in the same environment. However, whilst the computation time of the proposed method was comparable to some of the other mitigation methods, it was considerably slower than the Kalman filter. More work is needed to improve the computation time per frame of this method to ensure its viability for real-time interference mitigation.

VI. CONCLUSION

The data collected for this work demonstrates that interference is an issue for current automotive LiDARs. More than 1000 interference points were found in some point clouds, with measured ranges hundreds of metres greater than the range of the valid points.

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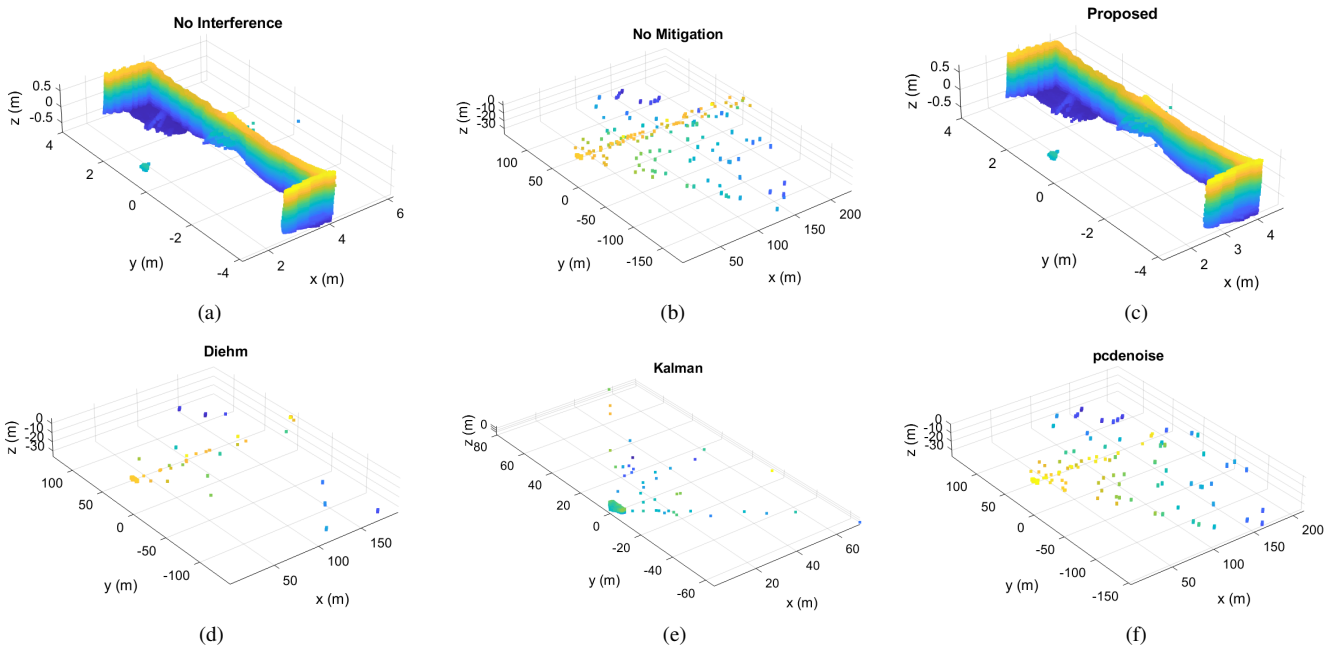

Fig. 2. Visualisation of Point Clouds for frame 10 from one of the data collections with 1m separation using a range threshold of 0.5m. (a) No Interference; (b) Interference without mitigation; (c) Proposed mitigation method; (d) Diehm mitigation method; (e) Kalman filtering; (f) pcdenoise mitigation method.

Consequently, this work presents a novel LiDAR mutual interference mitigation method that successfully suppresses interference from the first mitigated point cloud. The presented method only uses historical point clouds, so it can theoretically be used for real-time applications. To this aim, future work will explore the reduction of the computational time of the proposed method. The Kalman filter eventually produced stronger mitigation than the proposed method, but introduced noise in the initial mitigated point clouds. Hence, the proposed method is more suited to swiftly mitigate strong LiDAR mutual interference.

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